
AutoPolish: Improving Visual Aesthetics with Synthetic Audience Critique

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Abstract

1 Aesthetic judgment is plural. The same image can move one viewer and leave another cold, and both reactions are right, as they belong to different people. So the
2 correct answer to “how good is this image?” is not one number but the spread of
3 reactions an image draws from a crowd. We ask whether an off-the-shelf vision-
4 language model (VLM) can imitate that spread well enough to guide creative work.
5 We describe a single viewer persona in plain text, such as their age, education, and
6 taste in art, ask the model to react as that person would, and repeat for many view-
7 ers to build a synthetic audience and aggregate their reactions to form a simulated
8 crowd response. Across three datasets of human-rated images we show that only
9 19-47% of one person’s rating is shared with other viewers, thus an individual
10 score is mostly private taste and cannot be predicted. The crowd’s average, in
11 contrast, is steady enough that a fresh set of viewers reproduces it almost exactly
12 with 0.96 correlation. The synthetic audience also picks up how taste differs be-
13 tween kinds of viewers. When it predicts that one group will like an image more
14 than another group does, real people from those groups tend to agree, at a correla-
15 tion of 0.17, while without the viewer personas the correlation is only 0.02. This
16 carries over to AI-generated image pairs, where the simulated crowd can predict
17 which of two pictures people would prefer, and its accuracy gets better as more
18 viewer personas are added. Building on these findings, we introduce AutoPolish,
19 an auto-research style editing loop in which the synthetic audience serves as the
20 creative critic. It asks the simulated crowd to critique an image and turns their plu-
21 ral judgment into an improvement instruction for a language-conditioned image
22 editor. The edited image is then examined by a separate aesthetic judge, which de-
23 cides how successful the edit was and steers the next round of the loop. We show
24 that AutoPolish measurably improves aesthetic scores while keeping the picture
25 recognizably itself. We further show that the synthetic audience raises $3.1\times$ more
26 distinct objections than a single generic critic.
27

28 1 Introduction

29 Ask ten people to rate a photograph and you will rarely get one answer. A landscape that feels
30 serene to one viewer feels empty to another. An edit that one person calls striking another calls
31 garish. Editing tools usually treat this disagreement as noise and average it away [5, 11–13]. But an
32 editing signal built on a naive average ignores the plural nature of taste [2, 3] and tunes the result for
33 an average viewer who does not exist and who resembles no one.

34 Most work on computational aesthetics reduces the quality of a picture to a single score [5–7], and
35 thus the creative feedback loops spend their effort raising that one number [14, 15]. We take the
36 correct answer to “how will this image land?” to be the whole spread of reactions a picture draws

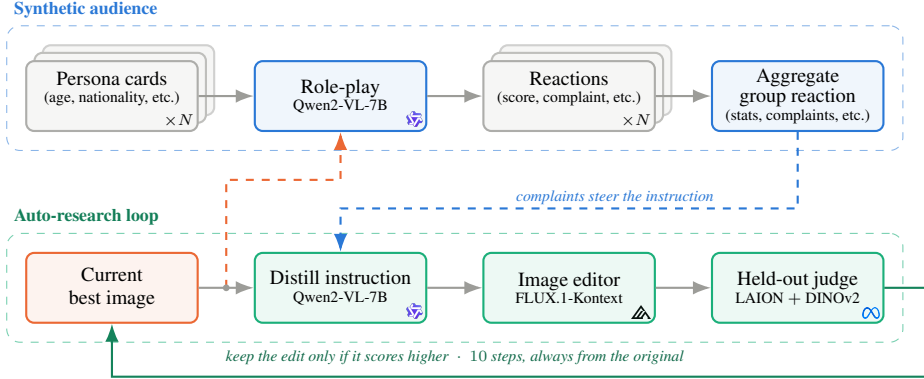


Figure 1: Overview of the AutoPolish pipeline. (Top) A frozen Qwen2-VL-7B is conditioned on N persona cards, each describing one viewer in plain text, to build a synthetic audience. Given an image, the aesthetic critique of every persona in that audience is aggregated and returned as a collective reaction. (Bottom) Given the current best image and the audience’s collective critique, Qwen2-VL-7B generates one editing instruction that resolves the critique. The instruction is fed to FLUX.1-Kontext, which produces a modified version of the image. A held-out aesthetic judge then examines the new image with a LAION aesthetic predictor and DINOv2, and decides whether the auto-research loop should roll back to the previous image or accept the new one.

37 from people. The feedback worth acting on is then where those opinions center, how widely they
 38 spread, and how they shift from one kind of viewer to another. The question this work addresses
 39 is whether an off-the-shelf vision-language model (VLM) can imitate that spread, and whether the
 40 imitation is trustworthy enough to steer a real creative decision.

41 Our method is deliberately straightforward and training-free; we describe a viewer in plain text by
 42 giving their age, education, familiarity with art, personality, and nationality. We then prompt the
 43 model to role-play that person and react to an image as they would, and we repeat this for many
 44 viewers to build a synthetic audience whose collective reactions we take as a prediction of the crowd
 45 response. The synthetic audience is then plugged into an auto-research style editing loop as the
 46 critic. It looks at the current picture and passes its complaints to a language-conditioned image
 47 editor as improvement instructions. A separate aesthetic judge decides whether the applied edit was
 48 beneficial without altering the essence of the picture. The judge’s decisions guide the iterations of
 49 the loop, either rolling back to the previous best image or moving forward with the proposed edit.

50 In short, our contributions are:

- 51 • We show that aesthetic taste is plural: as little as 19% of a rating is shared between individuals,
 52 while a group’s average is reproduced by a fresh set of viewers at up to 0.96 correlation.
- 53 • We show that a role-played synthetic audience is a usable proxy for a real crowd. It reproduces
 54 the differences in taste between viewer groups at a correlation $9\times$ that of a naive VLM.
- 55 • We show that adding the synthetic audience as the critic in an auto-research style editing loop
 56 draws $3\times$ more practical guidance than a generic critic and measurably improves aesthetic
 57 scores.

58 2 Related work

59 **Vision-language models as aesthetic judges.** Scorers built on CLIP made it practical to rate an
 60 image the model has never been trained on [11, 6], and instruction-tuned multimodal models im-
 61 proved on them by describing quality in words before turning it into a rating, which tracks human
 62 scores closely [24, 7]. They remain weaker than people at the finer judgments and at explaining
 63 them [25, 26], which is why models fine-tuned for aesthetics exist [27]. What all of them share is
 64 that they return one verdict per image, so they have nothing to say about how that verdict changes
 65 per viewer.

66 **Simulating Human Behavior with language models.** A separate literature asks models to answer
 67 as a particular kind of person. Given someone’s demographics, they reproduce survey answers with

surprising fidelity [1], though later work is more cautious: simulated answers vary less than real ones [28], they misrepresent several groups [29], and persona prompts slide easily into stereotype [30]. The lesson we carry over is that these simulations are trustworthy about groups rather than individuals, and only once calibrated. It is also a literature almost entirely about text, which leaves open whether a persona changes how a model sees a picture. That is the question we take up.

Feedback loops for editing. A third line of work refines outputs against feedback. Most of it uses a reward: a learned preference model [12, 13], reinforcement learning [31, 14], reward gradients [32, 33], or preference optimization [15]. A parallel line replaces the number with language, letting a model criticize its own output [34–36]; for images, a vision-language model finds the fault and an instruction-based editor repairs it [37–39, 10]. Two problems recur. A reward is a single number, so disagreement between viewers is averaged away before the model sees it, and when one model both proposes an edit and scores it, the improvement it reports is its own bias [40]. AutoPolish avoids both: its feedback comes from a group of viewers and arrives as words rather than a score, and the model that judges the result belongs to a different family from the one that makes the edit [11, 42].

3 Method

Persona cards. A persona card is a short textual description of a real annotator, generated deterministically from the attributes their dataset provides: age band, education, art and photography familiarity, Big-Five scores, and nationality. Attributes that a dataset does not provide are left out, not filled in with estimated values. A typical card reads “*Viewer profile: 35 to 44, master’s degree, high art familiarity, moderate photography experience, nationality: Belgium. Personality (Big-5, 1 to 5): O 4.2, C 3.1, E 2.8, A 3.9, N 2.5.*” We use the same template for the annotators we use from the Synthetic audience. Our claims are therefore limited to the attributes that the card actually contains.

The frozen LLM judge. The judge is a pretrained vision-language model that we query but never fine-tune. Each query supplies an image, a persona card, and an instruction to answer in a fixed JSON format; the model returns a score, a short rationale, and a few auxiliary fields. We parse it as strict JSON. If parsing fails, a per-field regular expression recovers the values, so we keep the sample instead of dropping it. The backbone is Qwen2-VL-7B-Instruct [9], in bf16. The persona-blind control format behaves as if no prompt conditioning were involved for the VLM judgment.

From a panel to a group prediction. To predict how a viewer group responds to an image, we sample N persona cards from the dataset and use them to judge the image, and summarize the reactions: the mean and spread of the scores, the most frequent complaints extracted from the rationales, and compare them with the ground truth scoring.

Averaging over the panel is not enough on its own, because the judge’s error is systematic and not random. Its ratings are pulled toward the middle of the scale: their standard deviation is only 51–58% of the human standard deviation, 34–56% of them land on a single value, and the endpoints are almost never used. Averaging cancels the random part of the error, but the compression is shared across the group population, so it survives. This is why raw population averages are less accurate than simply predicting the population mean. We therefore correct the scores after inference, using isotonic regression [43]. This fits a monotone mapping from the judge’s raw score to a calibrated one. We fit the mapping with two-fold cross-fitting over images, so every rating is calibrated by a mapping that was estimated without it. Because the mapping is monotone, it changes the absolute error but leaves every ranking as it was, and because it acts only on the output, the model weights stay frozen. The mapping also transfers across datasets. We can therefore fit it once on real images, where human ratings are available, and apply it unchanged to generated images, for which no per-image ratings exist.

AutoPolish. AutoPolish uses the group as the critic in an iterative editing loop, under the constraint that the model proposing edits and the model judging them are different, since a loop that scores its own edits can improve its objective without improving the image. Each image is refined for $R = 10$ steps of (i) **critique**: the group views the current image, and its complaints are aggregated across personas and accumulated across steps into a single instruction of at most fifteen words; (ii) **anchored re-edit**: FLUX.1-Kontext-dev [10] applies that instruction to the *original* image rather

than the previous edit, so editing artifacts do not compound, producing $K = 3$ candidates; (iii) **held-out scoring**: each candidate is scored by a LAION aesthetic predictor [11], a CLIP [41] backbone with an MLP head and thus a different model family from the critic, and checked for identity drift with DINOv2 [42]; (iv) **accept-if-better**: the new image replaces the previous best image if the aesthetic score improves and DINOv2 similarity to the original stays above 0.78, so the best-so-far score increases monotonically. We compare four critics: **STATIC**, a fixed “improve this image” instruction; **BLIND**, a single persona-free critique; **SOCIETY**, ten aggregated personas, which is our method; and **REWARD-ONLY**, which optimizes the aesthetic predictor directly and is an upper reference rather than a comparable baseline. Every condition appends the same suffix asking for a visible edit, so prompt wording cannot explain a difference between **SOCIETY** and **BLIND**, and we deliberately do not score instruction adherence, which would favour whichever condition writes the instruction that is easiest to follow.

4 Experimental setup

Data. We use three datasets of human-rated aesthetics. Each record pairs an image with the rating of one individual rater. **PARA** [4] contains 4,000 photographs and $\sim 52k$ ratings. It offers the richest persona fields: age, gender, education, art and photography experience, and Big-Five traits. **EVA** [18] consists of photographs taken from AVA [5] and provides 4,070 images and 137k ratings, but its persona fields are limited to age, gender, region, and photographic level. **LAPIS** [19] provides 4,000 paintings and 90k ratings, with nationality, art interest, and age. For generated content we use **Rapidata** [8], which contains 23,445 crowd votes on 898 pairs of AI-generated images. The raters come from 120 countries, and each vote records the rater’s country and language.

Protocol and metrics. We use the same judge in two versions: a persona-blind control, and a persona-conditioned one. Running many instances of the persona-conditioned judge, one per viewer, gives us the synthetic audience, and we vary how many instances we run. We always test on raters the model has never seen. What we mainly measure is whether the audience gets the *differences between groups* right: on one image, do the gaps we predict between two groups of viewers match the gaps the real raters show, and how does the size of the audience affect this? For AutoPolish we report how much the image improves, how much it still looks like the original, and how varied the critiques are. Every headline number carries a bootstrap 95% confidence interval.

5 Results

5.1 The ceiling: the group is learnable, the individual is not

Before asking how accurate a synthetic audience is, we should ask how accurate anything could be. One person’s rating is mostly personal: only 19-47% of it is shared with the other people who rated the same image, and the rest is theirs alone. A group average is a far steadier target; ask a second group of the same size and the two averages agree closely (0.84-0.96 across our datasets). Knowing who the rater is barely helps. Their listed attributes account for only 1-4% of their individual ratings, and a baseline that learns each person’s taste from their own past ratings gains almost nothing over ignoring them. This sets the bar for everything that follows: a synthetic audience should be judged on groups, and a method that claimed to predict one person’s taste accurately would be suspicious.

5.2 The synthetic audience is a faithful predictor of crowd reactions

The persona changes what the judge says. Giving the judge a persona moves its answers, and it moves them in the direction the real ratings go. On the two datasets that describe their raters in detail, the scores shift the right way ($r=0.37$ on **PARA**, 0.39 on **LAPIS**). The written comments shift even more: from the text alone we can tell what kind of viewer it was written for, well above chance on all three datasets, and the judge writes four times as many distinct comments as it does with no persona at all. On **EVA** this shows up only in the text, because the coarse integer score hides it. One correction is needed before any of this is usable. The judge rates almost everything near the middle of the scale, so we calibrate it once; this roughly halves the error of the audience’s average and puts it below simply guessing the average rating on every dataset (Figure 2, left).

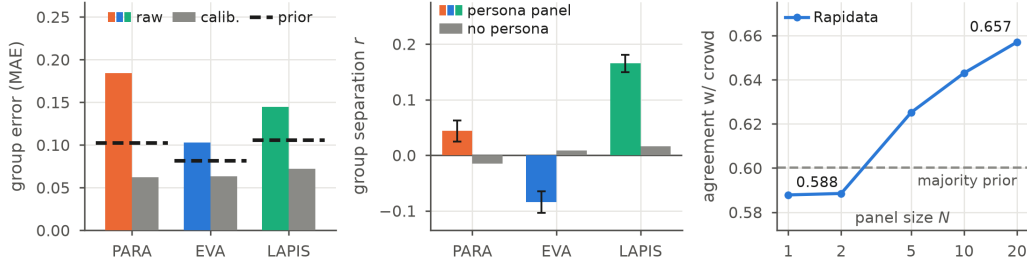


Figure 2: The synthetic audience is faithful to the real crowd reaction. (left) Calibration roughly halves group error and pushes the audience below the population-mean prior everywhere. (middle) The audience recovers real differences between viewer groups: the gaps it predicts between two groups on the same image track the gaps in the real ratings (bootstrap 95% CIs). (right) On AI-generated pairs, agreement with the crowd rises monotonically with the audience size and overtakes the majority prior.

Table 1: AutoPolish results. Every critic improves the image and every image stays recognizably itself, and AutoPolish does so while giving three times as much feedback per round as any other condition.

Critic	held-out gain (95% CI)	identity (final)	% improved	complaints/step
STATIC	+0.164 [0.135, 0.196]	0.968	84%	1.00
BLIND	+0.155 [0.126, 0.185]	0.964	72%	1.00
AUTOPOLISH (OURS)	+0.170 [0.136, 0.207]	0.951	69%	3.12
<i>reward-only (oracle)</i>	+0.187 [0.157, 0.221]	0.960	86%	n/a

The audience reproduces real differences between groups. Our main result is that the audience recovers not just the average opinion but the way opinion splits. On paintings, the gaps we predict between two groups of viewers looking at the same image line up with the gaps in the real ratings (+0.166, 95% CI [0.150, 0.181]), while the same judge without personas sits at essentially zero (+0.016; Figure 2, middle). It holds for each way of dividing the viewers on its own: by nationality (+0.068), by interest in art (+0.088), and by age (+0.102). On the photograph datasets the effect is weak but real on PARA (+0.044) and absent on EVA (−0.084). That ordering, LAPIS > PARA > EVA, is exactly the order of how much each dataset tells us about its raters.

The same mechanism works on AI-generated images. We repeat the test on 23,445 crowd votes over 898 pairs of generated images, asking the judge to answer as a voter from a given country and pick the better image of the two. A single simulated voter is close to a coin flip (0.532 correct). An audience of twenty agrees with the crowd 65.7% of the time, above the 60.0% reached by always choosing the more popular image, and agreement climbs steadily as the audience grows (Figure 2, right). This is the clearest view of why collective reaction works: each vote leans the right way only slightly, and averaging enough of them turns a slight lean into a reliable answer.

5.3 AutoPolish: a synthetic audience can steer image editing

We run AutoPolish on 100 photographs, 50 from PARA and 50 from EVA, with four critics and ten rounds each. The loop does what it is meant to do (Table 1). Every critic improves the image, measured by a scorer that plays no part in the editing, and the audience critic lifts the average aesthetic score from 5.10 to 5.27. The edited images still look like the originals, comfortably above the similarity floor of 0.78 we impose. Most of the improvement arrives within three or four rounds, and the condition that optimizes the scorer directly ends up only slightly ahead (Figure 3).

What sets the audience critic apart is what it says. It raises about three times as many distinct complaints per round as a single persona-free critic (3.12 versus 1.00), and it asks for concrete changes that a fixed instruction never suggests, such as plants and artwork in a bare office (Figure 4). The loop is easy to follow step by step: complaints accumulate, each round re-edits the original image rather than the previous edit, and the picture improves in visible increments (Figure 5).

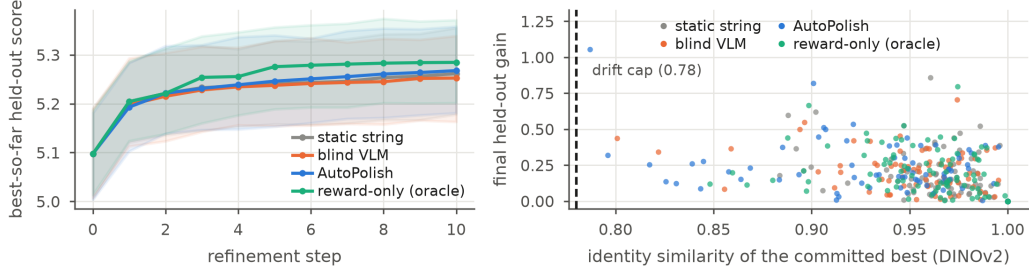


Figure 3: AutoPolish is fast and identity-preserving. (left) How the score of the best image so far rises with each round, averaged over all images (shaded band: 95% CI). Most of the gain arrives in the first three or four rounds. (right) Each dot showing how much the final image improved, against how much it still resembles the original. Almost every image sits far to the right of the similarity floor we impose.

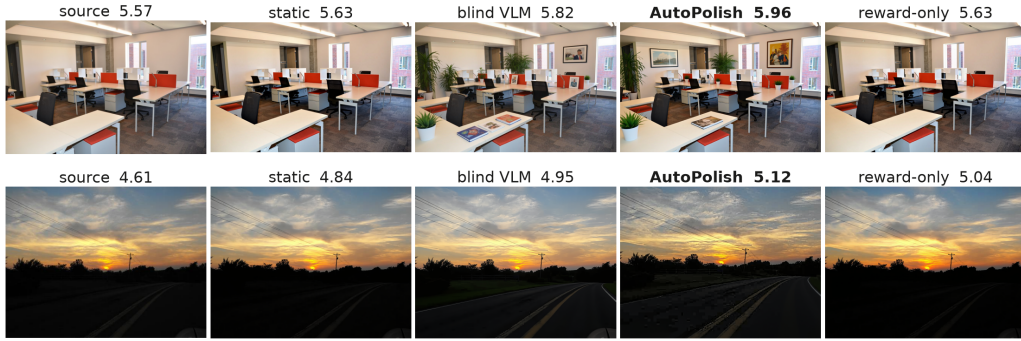


Figure 4: What the audience asks for. Source and each condition’s committed best edit, with held-out scores, for the images the audience critic helped most. AUTOPOLISH (bold) proposes semantically meaningful changes, such as plants and artwork in a bare office or light in a dark sunset, that STATIC and BLIND do not.

6 Conclusion

This work is ultimately about *agency*: who decides whether an image is good, and how much of that decision a creator hands over. Our premise is that there is no single verdict to hand over, because taste is plural. What we conclude is that this plurality can be simulated: a frozen model given personas, and listened to in numbers, is faithful to how a group would respond. And a reaction that can be simulated can be consulted while the work is still being made, which is what turns an audience into a critic. What that offers a creator is not a score but a voice they can hear and disagree with: the audience speaks, and the creator decides.



Figure 5: The loop over time. The committed best-so-far image after refinement steps 0-10 for the AUTOPOLISH critic, with the held-out aesthetic score above each cell. The audience’s complaints accumulate across steps, so the edits arrive as visible increments rather than one jump, and the score rises monotonically by construction.

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